INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

GRINDING MACHINE, INTERNAL, PLANETARY HEAD, HORIZONTAL SPINDLE, ELECTRIC MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, horizontal spindle, internal, planetary head grinding machines with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. The grinding machine shall be of the following styles. The style to be furnished shall be as specified (see 3.5 and 6.2).

Style A - Traversing wheelhead

Style B - Nontraversing wheelhead

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with

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specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B7.1 - Abrasive Wheels, Safety Requirements for the Use, Care, and Protection of. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identifications of Mounted Wheels. (DOD adopted)

ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

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ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - Tape, Perforated, One Inch. (DOD adopted)

RS-232 - Data Terminal Equipment and Data Communication Equipment, Interface Between, Employing Serial Binary Data Interchange. (DOD adopted)

RS-244 - Character Control For Numerical Machine Tool Control Perforated Tape. (DOD adopted)

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)

RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-491 - Interface Between a Numerical Control Unit and Peripheral Equipment Employing Asynchronous Binary Data Interchange Over Circuits Having RS-423-A Electrical Characteristics.

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, Standards Orders, 2001 Eye Street, N.W., Washington, DC 20006-1899.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The grinding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine linear axes position or tool movement shall be graduated in increments not larger than 0.0001 inch for the inch system, not larger than 0.002 mm for the metric system, or not larger than 0.0002 inch and 0.005 mm for the dual system of measurements. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system (DRO) shall be installed on the X, Y, and Z axes of travel. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system resolution shall be 0.0002 inch. The system accuracy shall be equal to its resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of $\pm$ 10 percent shall not adversely affect the operation of the system.

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3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis included within the DRO system. The feedback device shall be of independent type with a resolution equal to the resolution of the digital readout. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with a wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute and incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of the installed cables shall prevent any interference with normal machine operation.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured products are allowed under this specification.

3.2.3 Energy efficiency. The grinding machine and its components that directly conserve energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All mechanical and hydraulic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard

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on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA CFR 1910.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic lubrication system shall be provided for all parts and mechanisms requiring a continuous flow of lubrication. Means for determining lubrication level in each reservoir shall be by sight gauge, low level warning light, or similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. To provide for replacement or worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The grinding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the grinding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable " Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88, and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The grinding machine shall consist of basically the following components, as applicable to the style machine specified in 1.2. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. The base shall be a casting or stress relieved weldment, ribbed and braced to minimize deflection and vibration under maximum operating loads and to ensure accuracies required herein. When dust collectors or reservoirs for the coolant, or hydraulic systems are housed in the base, an access door(s) or removable plate(s) shall be provided. Means shall be provided for floor mounting and leveling of the machine.

3.4.2 Table. The table shall traverse longitudinally on guideways provided on the base. Means shall be provided to adjust and control the length and speed of table traverse. Unless otherwise specified (see 6.2), table traverse shall be hydraulically driven with means for manual operation. A safety device shall be provided to prevent damage due to table overtravel. The table on style B machines shall have precision ways to accommodate a cross slide.

3.4.3 Planetary head. The planetary head on style A machines shall be mounted on the traversing table. The planetary head on the style B machines shall be mounted on the base. The head shall house and support the inner and outer eccentrics, main drive spindle and its related components, and the grinding wheel assembly. The outer eccentric shall be driven by the main spindle drive motor. Means shall be provided for adjusting the relative position of inner and outer eccentrics to provide radial feed for the grinding wheel while the machine is in operation. The head shall be capable of accepting grinding spindles over a range length from 7-1/2 inches through 23

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inches. When specified (see 6.2), vertical adjustment of 0 to 7 inches shall be provided on style A machines.

3.4.4 Grinding wheel spindle. The grinding wheel spindle shall be mounted in precision antifriction bearings and shall be electric motor driven. The spindle shall rotate within the inner eccentric of the planetary head and shall orbit around the centerline of the surface being ground. Grinding wheel feed shall be accomplished by changing the orbital radius.

3.4.5 Cross slide. Style B machines shall have a cross slide. The cross slide shall traverse in a cross-axial direction mounted on the precision ways of the table. Positioning of the cross slide shall be accomplished manually by use of a graduated handwheel or other suitable means. When specified (see 6.2), the cross slide shall be power operated with means for manual control. The cross slide shall be adjustable vertically. All positioning screws and guides shall be protected from coolant, grinding dust, chips, and other foreign matter. Means for returning coolant to the reservoir shall be provided. Not less than two T-slots, configured in accordance with ANSI B5.1, shall be provided for mounting workparts and fixtures.

3.4.6 Auxiliary worktable. Unless otherwise specified (see 6.2), style A machines shall be provided with an auxiliary worktable. The worktable shall be separate and shall be constructed to bridge the base. Unless otherwise specified (see 6.2), the worktable shall have a sub-base constructed to accept a traveling cross slide for workpart positioning. The sub-base shall be an integral part of the machine base. Not less than three T-slots, configured in accordance with ANSI B5.1, shall be provided for mounting workparts and fixtures.

3.4.7 Wheel dresser. Unless otherwise specified (see 6.2), a manually operated wheel dresser, complete with mounting fixtures and the capability of dressing the entire periphery of the grinding wheel, shall be provided.

3.4.8 Grinding wheels. Grinding wheels shall be provided as specified (see 6.2). The wheels shall meet the requirements of ANSI B7.1 and ANSI B74.2 and shall be marked in accordance with ANSI B74.13.

3.4.9 Computer numerical control. When specified (see 6.2), the grinding machine shall be provided with a computer numerical control (CNC). When CNC is specified, the requirements in 3.4.9 through 3.4.9.9 shall apply. The control shall be completely programmable through a fully automatic, solid-state, soft-wired, integrated circuit-type computer numerical control. The control shall accept and translate program data from memory into positioning and feed motion command signals for the X and Z axes. Axes (X and Z) motion nomenclature shall conform to EIA RS-267. Control features shall include fixed cycles, control diagnostics, part program, and buffer edit capability, and part program storage. The machine control unit (MCU) shall accept part program data entered from a keyboard manual data input (MDI) and from an external input device. The control shall be provided with both EIA RS-232 and EIA RS-244 interface plugs for data input and output devices. A line voltage of $\pm$10 percent from normal shall not adversely affect the computer numerical control function. The control shall be capable of functioning in ambient temperatures ranging from 50° Fahrenheit (F) to 120°F and relative humidities ranging from 5 to 95 percent non-condensing. The

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control shall automatically shut down when internal operating temperature exceeds safe operating temperature. The control resolution shall be no greater than 0.0001 inch.

3.4.9.1 Data display. A cathode ray tube (CRT), capable of displaying at least 250 characters simultaneously, shall be provided. The control shall display the active and a minimum of the next two lines of information both in single block and automatic modes. The screen shall be automatically updated as each block of data is executed. Unless otherwise specified (see 6.2), the CRT shall, as a minimum, display the following data:

a. Mode of operation (i.e., automatic, single block, MDI, and manual).

b. Machine status (i.e., power on, in cycle, idle, end of cycle, in hold, and in E-stop).

c. Control status (i.e. manual spindle override, manual feedrate override, in edit, and program transfer active).

d. Actual and command position of each axis.

e. Program listing.

f. Control and system error messages.

3.4.9.2 External data input. The MCU shall accept part program data from an external input device compatible with the part program storage media (magnetic cassette, bubble cassette, diskette, magnetic card or punched tape) of the type specified (see 6.2). If punched tape is required the input and codes shall comply with EIA RS-227, EIA RS-274, and EIA RS-358.

3.4.9.3 Part program memory. The MCU shall be capable of storing in solid-state or bubble-type memory the contents of not less than 256k bytes of user accessible main memory.

3.4.9.4 Standby battery power. Standby battery power shall be provided for machines having volatile memory. The standby battery power shall be capable of retaining a fully loaded memory for a period of not less than 72 hours after the primary power source fails.

3.4.9.5 Part program edit. A program edit capability shall be provided to allow block insertion, additions, deletions, or modifications of a part program in the part program memory. Data related to part program edit shall be input by MDI provided on the control panel. Edited program memory shall be automatically acted upon, in the sequence designated, whenever the program is cycled.

3.4.9.6 Sequence number search. The control shall have the capability to permit the operator to search through the part program data for a designated block number by initiating the search through the use of the MDI keyboard. The search may be conducted through the tape data or stored part program data, depending on which is active.

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3.4.9.7 Block delete. The control shall have a delete feature to permit by-passing one or more instructions of programmed data.

3.4.9.8 Program data edit. Part program edit capability shall be provided as means for editing tape and MDI input to optimize part program data. It shall be possible to modify part program data by adding, deleting, correcting, or completely changing stored data and operating with the edited program.

3.4.9.9 Manual data input. A means for MDI shall be furnished to permit manual entry of alphabetic, numeric, and special characters necessary for control of all programmable machine functions in both EIA RS-491 and RS-274 format. The data shall be immediately displayed on the data display to permit data verification. The MDI shall permit entering a block of data into memory upon command.

3.4.10 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.11 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, and a provision for draining.

3.4.12 Electrical system. The electrical system shall be wired to conform to NFPA 79. The grinding machine shall draw all of its power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed for operation on 3-phase, 60-hertz (Hz), dual 230/460-voltage. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.12.1 Motors. Motors shall meet the requirements of 3.4.12. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.12.2 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield and an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.12.3 Hour meter. When specified (see 6.2), the grinding machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of

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the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by machine operation.

3.5 Style and capacity. Unless otherwise specified (see 6.2), the style and capacity of the machine shall meet the requirements of table I for the machine specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The machine shall grind workpieces in accordance with the test procedures given in 4.8.2 through 4.8.4.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished. Unless otherwise specified (see 6.2), the following additional equipment shall be furnished:

Ball bearing spindles, one each of the following sizes:
7-1/2", 10", 15", 18", 11" through 18", and 23 inches; spindles shall be complete with arms and collets.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

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3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE Number, Military, or Federal specification number)
Type of filter

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data shall be written in the English language.

3.13 Workmanship. Workmanship of the grinding machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance

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to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The grinding machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 50^{0} to 120^{0}F and 5 to 95 percent relative humidity non-condensing.

4.8.2 Operational test. The grinding machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

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4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The performance of each grinding machine shall be tested by grinding a bore not less than 3 inches diameter by 6 inches long in a suitable workpiece. Grinding wheels and materials of the test part shall be at the contractor's option. The ground bore shall be round within 0.0003 inch total indicator readout (TIR) and straight within 0.0002 inch over 6 inches length. Surface finish shall be not rougher than 8 microinches.

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grinding machines covered by this specification are intended for use in a machine shop environment where precision internal grinding is required. The grinder is designed primarily for use where non-symmetrical weight distribution, shape, or excessive size prevent work-piece rotation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

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e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Digital readout system, if not required (see 3.2.1.2).

g. Specify any special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Table traverse, if different (see 3.4.2).

m. Vertical adjustment on style A machines, if required (see 3.4.3).

n. Power operated cross slide, if required (see 3.4.5).

o. Auxiliary worktable on style A machines, if not required (see 3.4.6).

p. Auxiliary worktable construction, if different (see 3.4.6).

q. Wheel dresser, if different (see 3.4.7).

r. Grinding wheels, specify and fully describe (see 3.4.8).

s. Computer numerical control, if required (see 3.4.9).

t. Data display, if different (see 3.4.9.1).

u. External data input, specify and fully describe (see 3.4.9.2).

v. Electrical system, if different (see 3.4.12).

w. Initial voltage, if different (see 3.4.12).

x. Motors, if different (see 3.4.12.1).

y. Worklight, if not required (see 3.4.12.2).

z. Hour meter, if required (see 3.4.12.3).

aa. Style and capacity, if different (see 3.5).

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bb. Additional performance features, if required (see 3.7.1).

cc. Standard equipment, if different (see 3.8).

dd. Specify and fully describe optional equipment if required (see 3.9).

ee. Specify applicable repair parts, if required (see 3.10).

ff. Lubrication plate, if required (see 3.11.1).

gg. Information to be provided on lubrication plate, if different (see 3.11.1).

hh. Nameplate, if different (see 3.11.2).

ii. Hydraulic plate, if required (see 3.11.3).

jj. Hydraulic plate, if different (see 3.11.3).

kk. Inspection conditions, if different (see 4.3).

ll. First article inspection, if different (see 4.4).

mm. Quality conformance inspection, if different (see 4.5).

nn. Acceptance test, if different (see 4.6).

oo. Additional performance test(s), if required (see 4.8.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be the first article sample, a production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

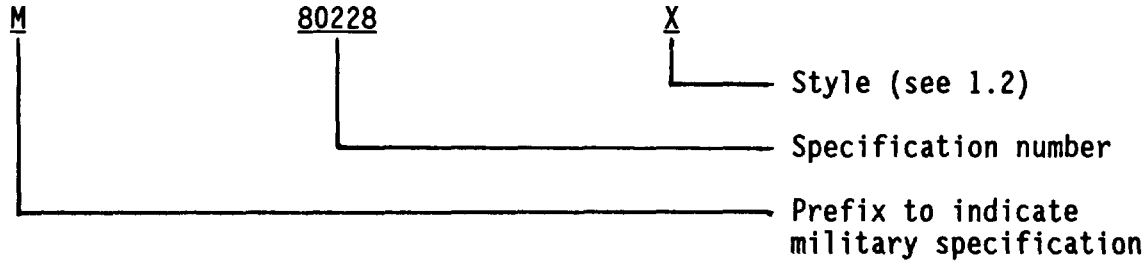
6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. requirements should be as specified by the procuring activity in the contract.

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6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Cylindrical
Equipment, industrial
Metalworking, precision
Shop, machine
Wheelhead

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TABLE I. Styles and capacities.

Characteristic	Requirement	
	Style A	Style B
Capacities		
*Work bore diameter, range, inches	1-1/8 to 14-1/2	1-1/8 to 6
*Grinding stroke, range, inches	0 to 34	0 to 23
Orbit radius, about grinding head center, range, inches	1-1/16 to 1-9/16	1/16 to 1-9/16
Cross slide travel adjustment, inches		
Vertical	-	5/16
Transverse	-	31
Cross slide table, length x width, inches		
Overall	-	48 x 17
Working surface	-	38 x 12-3/4
Table travel, inches	40	31
Table work surface to center of grinding diameter, inches	24	17-3/4
Auxiliary worktable, length x width, inches	76 x 28	-
Auxiliary worktable, cross slide travel, inches	26	-
Speeds (All listed speeds allowed a tolerance range ± 5%)		
Orbital motion speeds, number	variable	3
Revolutions per minute	0 to 55	50, 80, 110
Table feeds, number	variable	
Feet per minute	0 to 12	0 to 7
Motors, horsepower		
Wheel spindle	10	7-1/2
Orbital motion	5	**
Hydraulic	3	**
Coolant	3/4	1/4

*Obtained with proper assortment of grinding spindles.
**Utilizes wheel spindle motor.
Note: Machine capacities shall be not less than the stated requirements of the style shown in the above table, nor shall the machine offered exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE II. Alignment tolerances for styles A and B machines.

Test To Be Applied	Allowable Deviation
Bedways straight in longitudinal direction	0.0002"/12"
Bedways flat or level in longitudinal direction	0.0002"/12"
Workholding arrangement, cross slide (style B) or auxiliary worktable (style A), flat or level, measured at several positions	0.001"/12"
Table parallel with guideways (style B)	0.001"/12"
Pillow block parallel with guideways (style B)	0.001"/24"
Spindle parallel with guideways (style B)	0.001"/24"
T-slots in cross slide square with table guideways	0.001"/12"
Axis of wheel spindle parallel with worktable surface (style A)	0.0005"/12"
Spindle assembly parallel to longitudinal ways (style A)	0.001"/12"
Spindle assembly and longitudinal ways square to worktable (style A)	0.001"/12"
Rise and fall, cross slide, over full travel	0.001"/12"
Outer eccentric axial slip	0.0003"
Grinding wheel spindle radial run-out	0.0002" TIR
Grinding wheel spindle axial slip	0.0002"

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activity:
Navy - MC

Preparing activity:
DLA - IP

Project (3415-0179)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-G-80228B	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE: Grinding Machine, Internal, Planetary Head, Horizontal Spindle, Electrical Motor Driven

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander Defense Industrial Plant Equipment Center	b. TELEPHONE (Include Area Code) (1) Commercial: 901-775-4797 (2) AUTOVON: 683-4797
c. ADDRESS (Include Zip Code) ATTN: DIPEC-SSM Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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